



East West University
Department of Computer Science and Engineering
Course Outline
Semester: Summer 2026

Course: CSE325: Operating Systems (4 Credits), Section: 4,5,6

Credits and Teaching Scheme

	Theory	Laboratory	Total
Credits	3	1	4
Contact Hours	3 Hours/Week for 13 Weeks + Final Exam in the 14 th Week	2 Hours/Week for 13 Weeks	5 Hours/Week for 13 Weeks + Final Exam in the 14 th Week

❖ **Prerequisite :** None

❖ **Instructor Information**

Instructor: Shatabdi Roy Moon

Lecturer, Department of Computer Science and Engineering

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❖ **Course Objective**

The objective of this course is to enable students to acquire both theoretical foundations and practical competencies in operating system design and implementation. Students will examine the core components of operating systems, including process and thread management, scheduling, memory organization, inter-process communication, synchronization, deadlock handling, and file and storage management. Laboratory exercises and project work will reinforce conceptual understanding and provide hands-on experience with system-level programming. The course further establishes the prerequisite knowledge base for advanced courses in computer architecture and distributed systems.

❖ **Knowledge Profile**

K3 (Theory-based Engineering Fundamentals) A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline

K6 (Engineering practice (technology)) Knowledge of engineering practice (technology) in the practice areas in the engineering discipline

❖ **Learning Domains**

Cognitive - C3: Applying,
C4: Analyzing
Psychomotor - P2: Manipulation,
P3: Precision P4: Articulation
Affective - A2: Responding

❖ **Program Outcomes (POs)**

PO1: Engineering Knowledge
PO2: Problem Analysis
PO3: Design/Development of Solutions
PO5: Modern Tool Usage

❖ **Complex Engineering Problem Solution**

EP1: Depth of knowledge required
EP2: Range of conflicting requirements
EP3: Depth of analysis required

CO–PO Mapping

CO	CO Description	PO	Learning Domains	Knowledge Profile	CEP/ Activities
CO1	Explain and analyze the major components of an operating system in the context of providing an interface between hardware and users, to understand fundamental system design principles.	PO1	C3	K3	–
CO2	Apply and evaluate concepts of processes, threads, and scheduling within computing environments to manage resources efficiently and resolve performance trade-offs.	PO2	C3	K3	EP2
CO3	Analyze and justify algorithms for inter-process communication, synchronization, deadlock handling, memory management, and file/disk systems in real-life scenarios to design effective solutions for resource conflicts.	PO3	C4	K3	EP1, EP3
CO4	Design, implement, and document operating system solutions through laboratory experiments, and a mini-project in order to develop practical skills with modern system-level tools and practices.	PO5	C3, C4 P3, P4 A2	K6	EP1, EP2, EP3

Mid-Term Topics and Assessment (30%)

Course Topic	Teaching–Learning Method	CO	Cognitive Levels	Marks
Operating System Components	Lectures and discussions	CO1	C3	6
Processes	Lectures, exercises, discussions	CO1	C3	9

Threads	Lectures, tutorials, examples	CO2	C3	5
Scheduling Algorithms	Lectures, problem-solving	CO2	C3, C4	10

Final Exam Topics and Assessment (35%)

Course Topic	Teaching–Learning Method	CO	Cognitive Levels	Marks
Inter-Process Communication & Synchronization	Lectures, case studies, problem-solving	CO2	C4	10
Deadlock Handling	Lectures, exercises, problem-solving	CO3	C4	10
Memory Management	Lectures, tutorials, exercises	CO3	C4	10
File, I/O and Disk Management	Lectures, case studies	CO3	C4	5

Lab & Mini Project (CO4)

Assessment Area	CO	PO	Learning Domains	Knowledge Profile	CEP/Activities	Marks
Lab Performance	CO4	PO5	C3, C4 P3, P4 A2	K6	EP1, EP2	5
Lab Exam	CO4	PO5	C3, C4 P3, P4 A2	K6	EP1, EP2	10
Mini Project	CO4	PO5	C3, C4 P3, P4 A2	K6	EP1, EP2, EP3	5
Total						20

Marking Scheme for CO4 (20%)

Lab Performance & Lab Exam – 15%

Component	CO	PO	Learning Domains	Knowledge Profile	CEP/Activities	Marks
Weekly Lab Exercises (hands-on coding, implementation of OS concepts)	CO4	PO5	C3, P3	K6	EP1, EP2	4
Lab Reports / Documentation	CO4	PO5	C3, A2	K6	EP1	1
Lab Exam (individual, coding + viva)	CO4	PO5	C4, P4	K6	EP1, EP2, EP3	10
Total (Lab Performance)						15

Component	Description	Marks
Weekly Lab Exercises (Hands-on coding & implementation)	Completion and correctness of experiments (e.g., process management, threads, synchronization, memory, file/disk). Continuous assessment.	4
Lab Reports / Documentation	Clarity, completeness, and submission of reports for each lab.	1

Lab Exam (Individual)	One end-term lab exam testing practical implementation and OS concepts (coding, debugging, short viva).	10
Total		15

Mini Project Mapping (5%)

Component	CO	PO	Learning Domains	Knowledge Profile	CEP/Activities	Marks
Project Design & Implementation	CO4	PO5	C3, P3	K6	EP1, EP2	2
Report / Documentation	CO4	PO5	C3, A2	K6	EP1	1
Presentation / Viva	CO4	PO5	C3, P4, A2	K6	EP1, EP3	2
Total (Mini Project)						5

Component	Description	Marks
Project Design & Implementation	Correctness, functionality, and use of OS concepts in solving a moderately complex problem.	2
Report / Documentation	Clear explanation of objectives, methodology, results, and OS principles applied.	1
Presentation / Viva	Oral/poster presentation demonstrating understanding and teamwork.	2
Total		5

Overall Assessment Scheme

Assessment Area	CO1	CO2	CO3	CO4	Others	Total
Class Test / Quiz	05		05			10
Mid Term Exam (30%)	15	15				30
Final Exam (35%)		10	25			35
Class Performance					05	05
Lab Performance				05		05
Lab Exam				10		10
Mini Project				5		5
Total Marks	20	25	35	20	5	100

❖ Tentative Course Outline & Lecture Plan

SL.	Class no	Topics	Cognitive level	CO	Marks
1	Lecture - 1	Course Overview & OS Fundamentals: Components of the computer system, OS definition, polling v/s interrupt	C3	CO1	6
2	Lecture - 2	How Interrupt works, Types of Interrupts, DMA, User Communication with OS, Kernel & Shell			
3	Lecture - 3	GUI & CLI, Transition from User to Kernel Mode, Simple model for memory Protection, Kernel Types			

4	Lecture - 4	Operator Control, Batch Processing, Multiprogrammed Processing, Time-Shared Processing, Virtual Machines			
5	Lecture - 5	Program to Process, Process in memory, PCB, Two State Process model, Five states Process Model	C3	CO1	9
6	Lecture - 6	CPU Switch from process to process, Context Switch, Process scheduling, Schedulers, Process Creation.			
7	Lecture - 7	Process and Threads, Single and Multi-Threaded Processes, Threads Benefits, Threads v/s Processes	C3	CO2	5
8	Lecture – 8	Threads' Life cycle, Implementing threads: User Level, Kernel Level, Hybrid Level			
9	Lecture – 9	Multi-threading models: Many to One, One to One, Many to Many			
10	Lecture – 10	CPU-I/O Burst Cycle, CPU Scheduler, Preemptive v/s Non-Preemptive, Scheduling Criteria, The Perfect Scheduler	C3, C4	CO2	10
11	Lecture – 11	Scheduling Algorithms (Non-Preemptive): FCFS, SJF, Priority Scheduling, Convoy Effect			
12	Lecture – 12	Scheduling Algorithms (Preemptive): SRTF, Priority Scheduling, Round Robin			
13	Lecture – 13	Mid Term Assessment	-	-	-
14	Lecture – 14	Process Communication, Communication Models	C4	CO2	10
15	Lecture – 15	Producer-Consumer Problem, Race Condition, Critical Section			
16	Lecture – 16	Disabling Interrupts, Lock Variables, Strict Alternation, Peterson's solution			
17	Lecture – 17	Sleep & Wakeup, Semaphore: Binary & Counting			
18	Lecture – 18	Concept of Deadlock, RAG, Conditions for deadlock, Methods of handling Deadlocks	C4	CO3	10
19	Lecture – 19	Deadlock Prevention, Deadlock Avoidance, Safe & Unsafe State, Banker's Algorithm			
20	Lecture – 20	Memory Management: Multiprogramming, Swapping, Contiguous Allocation, Fixed Partitioning, Variable Partitioning			

21	Lecture – 21	Fragmentation, Compaction, First Fit, Best Fit, Next Fit, Worst Fit	C4	CO3	10
22	Lecture – 22	Paging: Architecture of Paging & Logical Address to physical address conversion, Segmentation: Architecture of Segmentation & Logical Address to physical address conversion			
23	Lecture – 23	Virtual Memory: FCFS, LRU & Optimal			
24	Lecture – 24	Concept of Disk Scheduling, Disk Architecture, Disk Scheduling algorithms: FCFS, SSTF	C4	CO3	5
25	Lecture – 25	Disk Scheduling algorithms: SCAN, C-SCAN, LOOK, C-LOOK			
26	Lecture – 26	Final Exam	-	-	-

❖ Grading System

Marks (%)	Letter Grade	Grade Point	Marks (%)	Letter Grade	Grade Point
80-100	A+	4.00	55-59	B-	2.75
75-79	A	3.75	50-54	C+	2.50
70-74	A-	3.50	45-49	C	2.25
65-69	B+	3.25	40-44	D	2.00
60-64	B	3.00	Below 40	F	0.00

❖ Exam Dates

Section	Mid Term Assessment	Final
4	TBA	TBA
5	TBA	TBA
6	TBA	TBA

❖ Academic Code of Conduct

Academic Integrity:

Any form of cheating, plagiarism, personification, or falsification of a document as well as any other form of dishonest behavior related to obtaining academic gain or the avoidance of evaluative exercises committed by a student is an academic offense under the Academic Code of Conduct and **may lead to severe penalties as decided by the Disciplinary Committee of the university.**

Special Instructions:

- Students are expected to attend all classes and examinations. A student **MUST** have at least 80% class attendance to sit for the final exam.
- Students will not be allowed to enter the classroom after 20 minutes of the starting time.
- For plagiarism, the grade will automatically become zero for that exam/assignment.
- Normally there will be **NO make-up exam**. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student misses any exam, the student **MUST** get approval for a makeup exam by written application to the Chairperson through the

Course Instructor **within 48 hours** of the exam time. Proper supporting documents in favor of the reason for missing the exam must be presented with the application.

- For the **final exam**, there will be NO makeup exam. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student misses the final exam, the student **MUST** get an approval of **Incomplete Grade** by written application to the Chairperson through the Course Instructor **within 48 hours** of the final exam time. Proper supporting documents in favor of the reason for missing the final exam must be presented with the application. **It is the responsibility of the student to arrange an Incomplete Exam within the deadline mentioned in the Academic Calendar in consultation with the Course Instructor.**
- All mobile phones and smart watches **MUST** be turned to silent mode during class and exam periods. Students cannot carry any Bluetooth device in the exam hall.
- There is **zero tolerance for cheating** in exams. Students caught with cheat sheets in their possession, whether used or not; writing on the palm, back of calculators, chairs, or nearby walls; copying from cheat sheets or other cheat sources; copying from other examinee etc. would be treated as cheating in the exam hall. The only penalty for cheating is **expulsion for several semesters as decided by the Disciplinary Committee of the university.**